Nirbhay Singh

FinOps & Cloud Cost Optimisation Architect | EUR 4M+ Documented Savings

EUR 4M+	$45K	60%	50+	14+
Total Savings	AWS Savings (Deloitte)	AI Cost Reduction	Accounts Governed	Certifications

PROFESSIONAL SUMMARY

Cloud & FinOps Architect with 11+ years of experience and EUR 4M+ in documented cloud cost savings at Bosch Europe. Expert in FinOps governance, cloud cost optimisation across AWS, GCP, and Azure, and building open-source cost intelligence tools. Created TokenMeter (LLM cost tracker achieving 60% AI spend reduction) and InfraCents (Terraform cost estimation on every PR). Combines deep cloud architecture expertise with strategic cost governance -- helping organisations maximise cloud ROI through data-driven decisions, cultural change, and automated cost controls without sacrificing performance, security, or innovation velocity.

KEY ACHIEVEMENTS

- Delivered EUR 4M+ in documented cloud savings at Bosch Europe through FinOps governance, optimisation, and cost-aware architecture
- Reduced AWS costs by $45,000 at Deloitte through Spot Instance fleet management, Reserved Instances, workload right-sizing, and pipeline efficiency
- Built TokenMeter (open-source): LLM cost intelligence layer providing per-feature/per-team breakdowns with smart routing saving 60% on AI spend
- Built InfraCents (open-source): GitHub App posting real-time cloud cost estimates (AWS+GCP) on every Terraform PR before merge
- Established enterprise FinOps practice: chargeback/showback models, tagging strategies, budget automation, anomaly detection, and executive reporting
- Optimised AI/ML workload costs through GPU scheduling, preemptible instances, Spot Instance strategies, and model right-sizing
- Led FinOps culture transformation -- trained 50+ engineers on cost-conscious architecture and cloud spending accountability

PROFESSIONAL EXPERIENCE

Cloud & AI Architect (FinOps Lead)

Dec 2022 - Present

Robert Bosch (Bosch Polska) Warsaw, Poland

- Implemented enterprise FinOps programme at Bosch Europe, delivering EUR 4M+ in documented savings through optimisation, governance, and cost-aware architecture
- Established chargeback/showback models, tagging strategies, and automated budget alerts across 50+ AWS and GCP accounts serving 15+ business units
- Built real-time cost dashboards and anomaly detection systems providing cloud spend visibility to engineering and executive stakeholders
- Defined resource lifecycle policies, right-sizing cadence, and reserved/committed capacity purchasing strategies based on usage analysis
- Led FinOps culture transformation: trained 50+ engineers on cost-conscious architecture, established cost reviews in architecture governance
- Optimised AI/ML workload costs on Vertex AI and GKE through GPU scheduling, preemptible instances, and workload bin-packing strategies

Senior DevOps Consultant

Jun 2021 - Nov 2022

McAfee

- Automated infrastructure with Terraform IaC, eliminating resource waste from manual provisioning and reducing deployment overhead by 50%+
- Implemented cost-aware container orchestration across GKE, EKS, and AKS with cluster auto-scaling, pod resource limits, and spot integration
- Built monitoring dashboards correlating infrastructure costs with performance metrics for data-driven optimisation decisions
- Identified and eliminated $100K+ in unused resources through automated cleanup scripts and resource lifecycle policies

Nirbhay Singh

FinOps | Cloud Cost Governance
Savings Plans | Right-Sizing | AI Cost Mgmt

CONTACT

@ nirbhaysingh1@live.com
in linkedin.com/in/nirbhaysingh1
GH github.com/nirbhays
M medium.com/@nirbhaysingh1
W cloudtoai.in
C credly.com/users/nirbhaysingh
L Warsaw, Poland

CORE SKILLS

FINOPS

FinOps Framework, Cloud Cost Allocation, Chargeback/Showback, Unit Economics, Cloud Billing APIs

COST OPTIMISATION

Spot/Preemptible, Reserved/Committed, Savings Plans, Right-Sizing, Auto-Scaling, Resource Lifecycle

CLOUD BILLING

AWS (Cost Explorer, CUR, Budgets), GCP (Billing, BQ export), Azure (Cost Mgmt)

COST TOOLING

TokenMeter, InfraCents, Infracost, Kubecost, CloudHealth

INFRASTRUCTURE

Terraform, Kubernetes, Docker, CI/CD, GitHub Actions, Jenkins

DATA & DASHBOARDS

BigQuery, Redshift, Grafana, Looker, Python, SQL

GOVERNANCE

Tagging, Multi-Account, Budget Alerts, Policy-as-Code, OPA

EDUCATION

MTech (Master of Technology)

Birla Institute of Technology

LANGUAGES

English	Professional
Hindi	Native

Senior DevOps Engineer / Data & AI Architect

Dec 2018 - Jun 2021

Deloitte

- Reduced AWS costs by $45,000 through Spot Instance fleet management, Reserved Instances, Savings Plans, and workload right-sizing
- Orchestrated 1,000+ containers using AWS Batch with Spot Instances -- maximised cost efficiency for distributed batch processing at 70% compute savings
- Accelerated CI/CD pipeline speed by 30x, reducing build/test compute costs and improving developer productivity
- Implemented cost monitoring, anomaly detection, and budget alerting for multi-cloud environments (AWS, GCP, Azure, OCI)

Senior Cloud Infrastructure Engineer

Nov 2015 - Dec 2018

PTC

- Designed cost-optimised AWS architectures: Auto Scaling policies, Spot Instances for non-critical workloads, Reserved Instances for baseline
- Managed 11,000+ server infrastructure -- identified and executed consolidation opportunities saving significant operational costs
- Led cloud migration planning with detailed TCO analysis, cost projection models, and ROI justification for executive stakeholders

System Engineer

Mar 2013 - Jul 2014

IBM

- Infrastructure operations and capacity planning for production environments -- early exposure to cost/capacity optimisation

OPEN-SOURCE PROJECTS

TokenMeter -- Cost Intelligence Layer for LLM Apps

>> github.com/nirbhays/tokenmeter

One-line integration tracking every token, cost, and latency across OpenAI, Anthropic, Google. Per-feature and per-team breakdowns. Smart routing saves up to 60% by matching task complexity to model capability. <5ms overhead.

STACK: Python, OpenAI API, Anthropic API, Google AI, Smart Routing, PyPI

InfraCents -- Terraform Cost Estimates on Every PR

>> github.com/nirbhays/infracents

GitHub App posting real-time AWS+GCP cost estimates on Terraform PRs before merge. Engineers see exact cost impact of infrastructure changes. Zero config, live pricing APIs.

STACK: Python, Next.js, Terraform, GitHub App, AWS/GCP Pricing APIs

ShieldIaC -- AI-Powered IaC Security Scanner

>> github.com/nirbhays/shieldiac

Prevents costly security incidents by catching IaC misconfigs pre-production. 100+ rules, 9 compliance frameworks, AI fix suggestions.

STACK: Python, Terraform, CloudFormation, GPT-4.1, GitHub Actions

CERTIFICATIONS & CREDENTIALS

- **AWS** Solutions Architect - Professional (Feb 2020)
- **AWS** Solutions Architect - Associate (Jan 2021)
- **Google** Professional Cloud Architect (Valid Jun 2026)
- **Google** Professional Cloud DevOps Engineer (Valid Dec 2027)
- **Google** Professional ML Engineer (Valid Dec 2027)
- **Google** Generative AI Leader (Valid Aug 2028)
- **HashiCorp** Terraform Associate
- **CKAD** - Certified Kubernetes Application Developer (Valid Sep 2026)

TECHNICAL WRITING & THOUGHT LEADERSHIP

- "Our AI Bill Was $4,800 Last Month -- Nobody Knew Why"
  How engineering teams can reclaim control of LLM spend: per-feature cost breakdowns, smart model routing, and the story behind building TokenMeter.
  https://medium.com/@nirbhaysingh1